SECURITY EAGLES

eJPTv2

Junior Penetration Tester Certification

STUDY GUIDE

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FROM LEARNERS TO LEADERS

Network Attacks

eJPTv2 Certification Study Guide

From Learners to Leaders

Security Eagles

January 27, 2026

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7 Network Attacks

Network attacks target the communication layer of systems. This section covers traffic analysis, network enumeration techniques, and protocol-specific attacks that are critical for the eJPTv2 certification.

7.1 Wireshark Basics & Traffic Analysis

Wireshark is the industry-standard network packet analyzer, allowing you to capture and inspect network traffic in detail.

7.1.1 Wireshark Fundamentals

What is Wireshark?

Wireshark captures packets traveling through your network interface and displays them in human-readable format. This is essential for understanding network communication, detecting security issues, and discovering sensitive data transmitted in plaintext.

Key Capabilities:

- **Live Capture:** Real-time packet monitoring from active interfaces
- **Protocol Analysis:** Deep packet inspection across OSI layers
- **Filtering:** Advanced display and capture filters for focused analysis
- **Troubleshooting:** Diagnose connectivity and performance issues
- **Forensics:** Extract and reconstruct files and conversations
- **Security Analysis:** Identify suspicious traffic patterns
- **Data Extraction:** Recover credentials, cookies, and sensitive information

Why Wireshark Matters for eJPTv2:

Many vulnerabilities involve unencrypted network protocols. Wireshark reveals:

- HTTP credentials and session cookies
- FTP/Telnet plaintext authentication
- DNS queries revealing internal infrastructure
- SNMP community strings and OIDs
- SMB credentials and file transfers

Installation and Setup

[EXAMPLE] Installing Wireshark

```
# Linux installation
sudo apt-get install wireshark

# Kali Linux (pre-installed)
wireshark &

# Allow non-root capture (optional)
sudo usermod -aG wireshark $USER
sudo chmod +x /usr/bin/dumpcap
```

7.1.2 Capturing Packets

Starting a Packet Capture

[EXAMPLE] Basic Packet Capture

1. Launch Wireshark
2. Select network interface (eth0, wlan0, etc.)
3. Click blue shark fin icon to start capture
4. Wireshark displays packets in real-time
5. Click red square to stop capture

[TIP] Choosing the Right Interface

Common Network Interfaces:

- **eth0/eth1:** Ethernet adapters
- **wlan0:** Wireless adapter (WiFi)
- **lo:** Loopback interface (localhost traffic only)
- **tun0:** VPN or tunneling interface

View available interfaces:

```
ifconfig
ip addr show
```

7.1.3 Wireshark Interface and Examples

Wireshark Main Interface - Packet Capture in Action

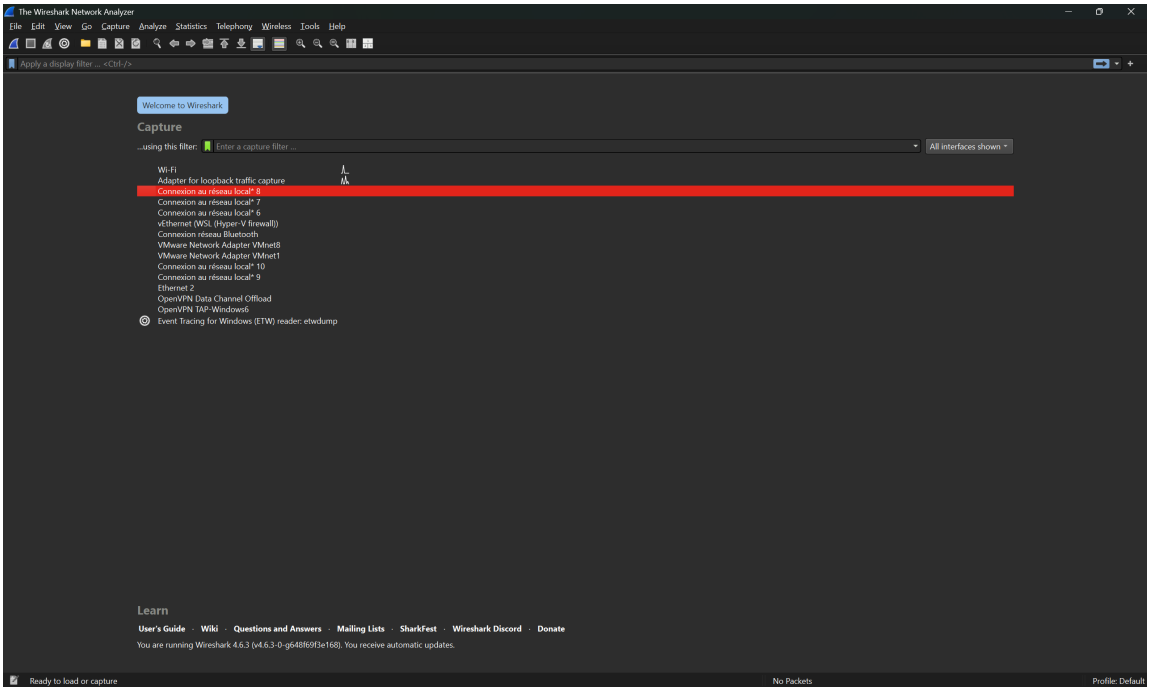

Figure 1: Wireshark main interface showing live packet capture with packet list, details pane, and hex viewer

Analyzing Captured Packets

Protocol-Specific Analysis

Network Traffic Filtering and Statistics

Extracting Data from Packets

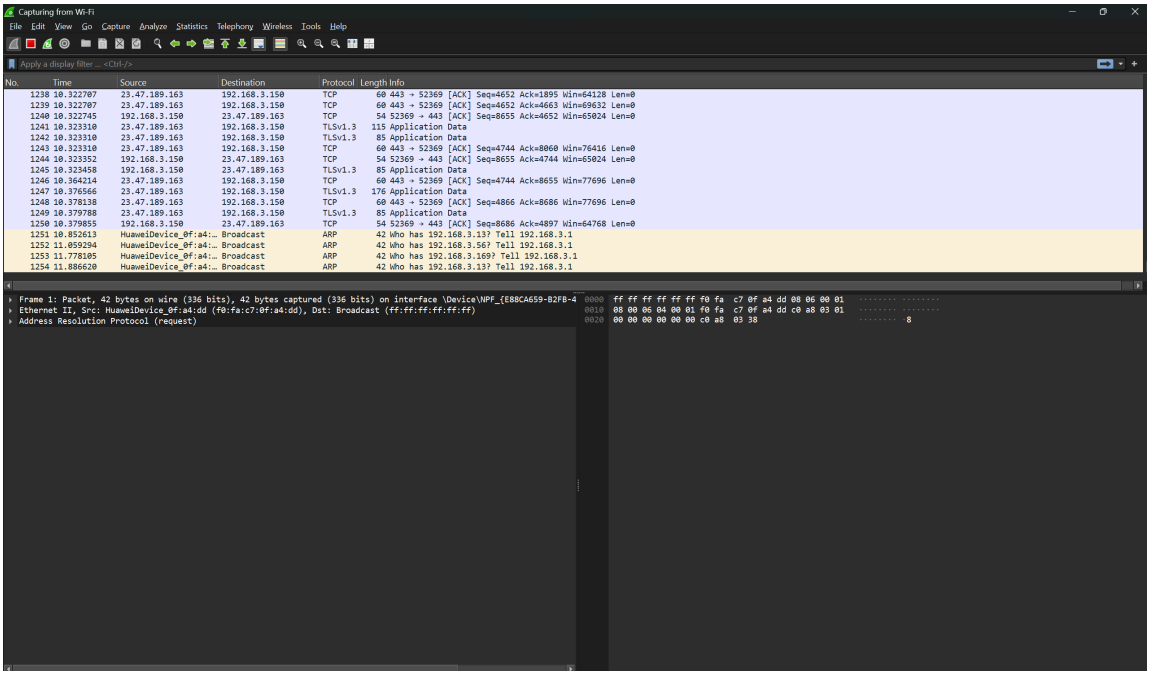

Figure 2: Detailed packet analysis showing protocol breakdown and packet layers

Capture Filters

Capture filters are applied DURING capture to reduce noise:

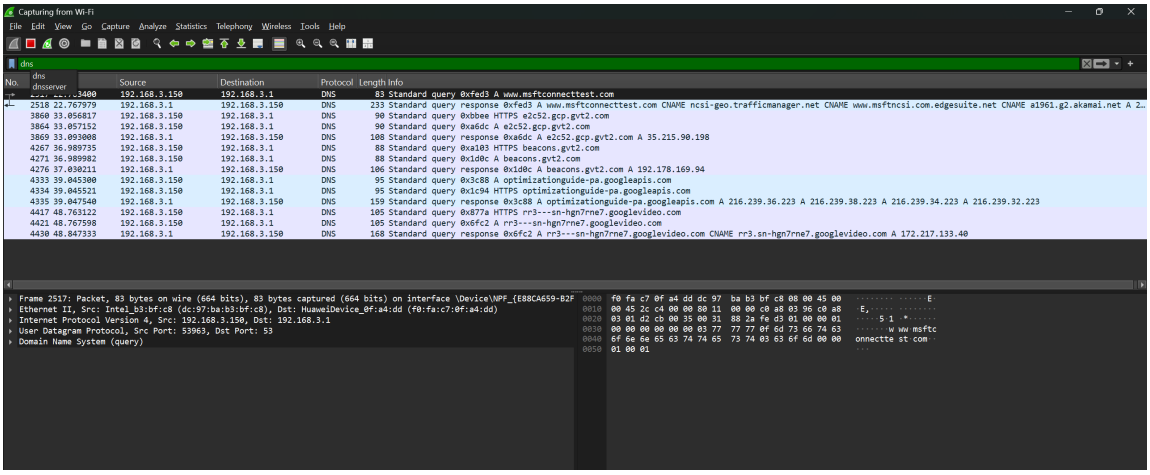

Figure 3: Wireshark display showing HTTP protocol analysis with request/response details

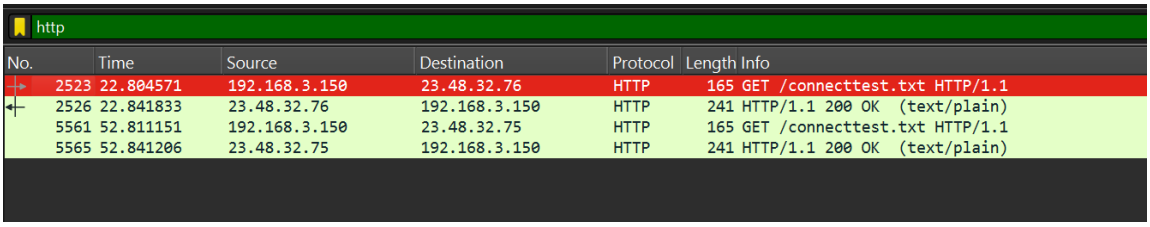

Figure 4: Filtered network traffic view with specific protocol isolation

[EXAMPLE] Useful Capture Filters

```
# Capture only HTTP traffic
tcp port 80

# Capture traffic to/from specific IP
host 192.168.1.100

# Capture SSH traffic
tcp port 22

# Capture FTP traffic
tcp port 21

# Capture traffic from specific source
src host 192.168.1.50

# Capture traffic to specific destination
dst host 8.8.8.8

# Combine filters with AND/OR
tcp port 80 or tcp port 443

# Exclude traffic
not icmp
```

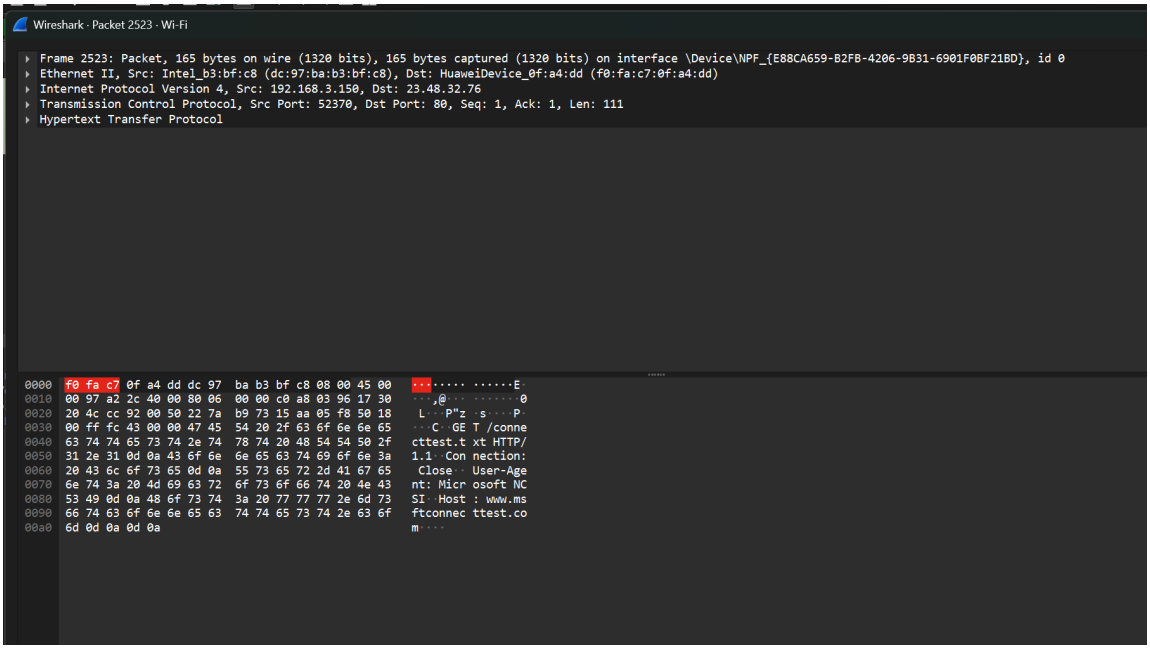

Figure 5: Wireshark showing data extraction capabilities for exported files and credentials

Display Filters

Display filters are applied AFTER capture to filter viewing:

[EXAMPLE] Display Filters

```
# Filter by protocol
http
dns
ftp
smb

# Filter by IP address
ip.src == 192.168.1.100
ip.dst == 8.8.8.8

# Filter by port
tcp.port == 22
tcp.dstport == 3306

# Filter by protocol and port
tcp.port == 80 and http

# Filter out noise
!arp and !dns

# Find specific strings
http.request.uri contains "login"

# Combine filters
(tcp.dstport == 80 or tcp.dstport == 443) and ip.src == 192.168.1.50
```

7.1.4 Analyzing Packet Details

Packet Structure

Each packet has three main sections:

1. **Packet List:** Shows all captured packets with summary
2. **Packet Details:** Detailed breakdown of selected packet layers
3. **Packet Bytes:** Raw hexadecimal and ASCII representation

[EXAMPLE] Understanding Packet Layers

HTTP GET Request breakdown:

- **Frame:** Physical layer information
- **Ethernet:** MAC addresses (Layer 2)
- **IP:** Source and destination IP (Layer 3)
- **TCP:** Source and destination ports (Layer 4)
- **HTTP:** Request method, URL, headers (Layer 7)

Following TCP Streams

TCP stream follows reconstructs the complete communication between two hosts:

[EXAMPLE] TCP Stream Analysis

1. Right-click on TCP packet
2. Select "Follow -> TCP Stream"
3. Wireshark displays client (blue) and server (red) communication
4. Can export conversation as file

7.1.5 Extracting Data from Captures

Exporting Objects

[EXAMPLE] Extracting Files from Captures

Export HTTP objects:

1. File -> Export Objects -> HTTP
2. Select files (images, documents, etc.)
3. Save to disk

Export other protocols:

- File -> Export Objects -> SMB
- File -> Export Objects -> TFTP

Password Extraction

[WARNING] Extracting Credentials

Many protocols transmit credentials in plaintext over the network:

- **HTTP:** Basic authentication headers
- **FTP:** Login credentials in traffic
- **TELNET:** All traffic is plaintext
- **IMAP/POP3:** Email credentials

Analysis Method:

```
# Filter for specific protocol
ftp

# Look for USER and PASS commands
# Credentials visible in packet details
```

7.1.6 Common Protocols in Wireshark

Protocol	Port	Analysis Focus
HTTP	80	URLs, headers, cookies, plaintext data
HTTPS	443	Encrypted (see SNI for domain)
DNS	53	Domain queries and responses
FTP	21	Credentials, file transfers
SSH	22	Encrypted session
SMB	445	File shares, authentication
TELNET	23	Plaintext credentials
SMTP	25	Email transmission

7.1.7 Practical Wireshark Examples

Capturing HTTP Traffic

[EXAMPLE] HTTP Traffic Analysis

Scenario: User logs into a website with plaintext password
Wireshark Filter:

```
http.request.method == "POST"
```

Analysis:

- Look at HTTP packets
- Expand HTTP layer in packet details
- View Form Data section
- Extract username and password fields

Capturing DNS Queries

[EXAMPLE] DNS Traffic Analysis

Wireshark Filter:

```
dns
```

Information revealed:

- Domains being queried
- DNS servers used (recursive vs. authoritative)
- DNS response answers
- NXDOMAIN (domain not found) responses

[OUTPUT] Terminal Output: DNS Query Example

```
Frame 142: 73 bytes on wire
Ethernet II: Src (aa:bb:cc:dd:ee:ff), Dst (ff:ff:ff:ff:ff:ff)
IPv4: 192.168.1.50 > 192.168.1.1
UDP: Src Port 54321, Dst Port 53
DNS: Standard query 0x1234 A example.com
    [Response In: 143]
    Queries
        example.com: type A, class IN
```

7.1.8 Network Protocol Fundamentals

OSI Model and TCP/IP Stack

The network protocol stack defines how data flows through network layers:

Layer	OSI	TCP/IP	Protocol Examples	Wireshark View
Application	5-7	Application	HTTP, HTTPS, FTP, SSH, SMTP, DNS	Decoded data, headers
Transport	4	Transport	TCP, UDP, SCTP	Port numbers, segments
Internet	3	Internet	IP, ICMP, IGMP	IP addresses, routing
Data Link	2	Link	Ethernet, ARP	MAC addresses, frames
Physical	1	Physical	Cables, signals	Bit-level data

[TIP] Understanding Protocol Layers

When analyzing packets in Wireshark, you'll see the protocol stack layered from bottom (Physical) to top (Application). Each layer adds its own header information (encapsulation). To extract data from a specific layer, click the expand arrows in the packet details pane.

Common Network Communication Flows

TCP Three-Way Handshake (Connection Establishment):

1. **SYN:** Client sends synchronization request (flags: SYN)
2. **SYN-ACK:** Server acknowledges and sends its own sequence (flags: SYN, ACK)
3. **ACK:** Client acknowledges server response (flags: ACK)
4. **Connection established** - Data can now be transmitted

In Wireshark, filter for handshakes with: `tcp.flags.syn==1 or tcp.flags.ack==1`

UDP Communication (Connectionless):

- Client sends data directly without handshake
- Server receives and optionally responds
- No acknowledgment or connection state tracking
- Faster but no reliability guarantees

Attack Flow Diagram - Network Reconnaissance

Typical penetration tester workflow:

Reconnaissance Phase
1. Passive Information Gathering (OSINT, DNS queries, network scanning)
2. Active Network Scanning (Ping sweep, ARP scan, port discovery)
3. Service Enumeration (Version detection, banner grabbing)
4. Vulnerability Analysis (Cross-reference with exploit databases)
5. Exploitation Planning (Select targets, test payloads)

7.2 Network Enumeration

Network enumeration is the process of systematically discovering and mapping network resources, services, and devices.

7.2.1 Network Enumeration Methodology

Enumeration Phases

1. **Network Discovery:** Identify live hosts
2. **Port Scanning:** Discover open services
3. **Service Identification:** Determine service versions
4. **Vulnerability Assessment:** Identify weaknesses
5. **Documentation:** Record findings

7.2.2 Host Discovery

Ping Sweep

[EXAMPLE] ICMP Ping Sweep

```
# Using ping
for i in {1..254}; do ping -c 1 192.168.1.$i | grep "bytes from"; done

# Using nmap (faster)
nmap -sn 192.168.1.0/24

# Identifying live hosts
nmap -sn 192.168.1.0/24 -oG - | grep "Up"
```

[OUTPUT] Terminal Output: Nmap Host Discovery

```
root@kali:~# nmap -sn 192.168.1.0/24

Starting Nmap at Mon Jan 12 14:00:00 2024
Nmap scan report for 192.168.1.1
Host is up (0.0012s latency).
Nmap scan report for 192.168.1.50
Host is up (0.0034s latency).
Nmap scan report for 192.168.1.100
Host is up (0.0021s latency).
Nmap scan report for 192.168.1.150
Host is up (0.0045s latency).
Nmap done at Mon Jan 12 14:01:00 2024; 4 IP addresses scanned, 4 hosts up
```

ARP Scanning

[EXAMPLE] ARP-based Network Scanning

```
# Using arp-scan (very fast)
arp-scan -l

# Using netdiscover
netdiscover -r 192.168.1.0/24

# Nmap with ARP
nmap -sn 192.168.1.0/24 --send-eth
```

[OUTPUT] Terminal Output: ARP Scan Results

```
root@kali:~# arp-scan -l

Interface: eth0, datalink type: EN10MB (Ethernet)
Starting arp-scan 1.10.0 with 256 hosts (https://github.com/royhills/arp-
    scan)
192.168.1.1      00:11:22:33:44:55       NETGEAR
192.168.1.50     aa:bb:cc:dd:ee:ff       Apple
192.168.1.100    de:ad:be:ef:00:01       (Unknown)
192.168.1.150    ca:fe:ba:be:00:02       Intel Corp

4 packets transmitted, 4 packets received,   0% packet loss
Scan time: 0.105 seconds
```

7.2.3 Port and Service Enumeration

Comprehensive Port Scanning

[EXAMPLE] Nmap Port Scanning Strategies

```
# Quick scan of top ports
nmap -sV 192.168.1.100

# All ports with version detection
nmap -p- -sV 192.168.1.100

# Intense scan
nmap -A -T4 192.168.1.100

# OS detection and script scanning
nmap -O -sC 192.168.1.100

# UDP port scanning
nmap -sU -p 53,123,161 192.168.1.100

# Aggressive scan (includes traceroute)
nmap -A -T4 --traceroute 192.168.1.0/24
```

Service Version Detection

[EXAMPLE] Version Detection

```
# Service version detection (-sV)
nmap -sV --version-intensity 9 192.168.1.100

# Probe open ports only
nmap -sV --version-light 192.168.1.100

# Combined with script scanning
nmap -sV -sC 192.168.1.100
```

7.2.4 DNS Enumeration

DNS Zone Transfer

[EXAMPLE] DNS Zone Transfer Attack

```
# Attempt zone transfer with dig
dig @192.168.1.1 example.com axfr

# Using nslookup
nslookup
> server 192.168.1.1
> set type=NS
> example.com
> set type=A
> example.com

# Using host command
host -t axfr example.com 192.168.1.1
```

[WARNING] Zone Transfer Risks

DNS zone transfers can leak internal network topology, server names, and IP addresses. This is a serious information disclosure vulnerability.
Mitigation:

- Restrict zone transfers to authorized servers only
- Implement DNSSEC
- Use access lists on DNS servers

DNS Record Enumeration

[EXAMPLE] Enumerating DNS Records

```
# A records (IPv4)
dig example.com A

# MX records (Mail servers)
dig example.com MX

# NS records (Nameservers)
dig example.com NS

# TXT records (can contain SPF, DKIM, etc.)
dig example.com TXT

# All records
dig example.com ANY

# Reverse DNS lookup
dig -x 8.8.8.8
```

7.3 SMB & NetBIOS Enumeration

SMB (Server Message Block) is a network file sharing protocol commonly used in Windows environments. NetBIOS provides name resolution on local networks.

7.3.1 SMB Overview

SMB Ports

- **Port 139:** NetBIOS Session Service (older)
- **Port 445:** Direct SMB over TCP (modern)

SMB Versions

Version	Characteristics
SMB1	Legacy, vulnerable (EternalBlue, etc.)
SMB2	Introduced in Windows Vista/2008
SMB3	Current, with encryption support

SMB Vulnerability Timeline

Vulnerability	Affects	CVE	Impact
EternalBlue	SMB1	CVE-2017-0144	RCE, WannaCry ransomware
EternalRomance	SMB1	CVE-2017-0145	RCE, lateral movement
ZeroLogon	NetLogon	CVE-2020-1472	Privilege escalation
SMB Null Session	All versions	N/A	Unauthenticated enumeration
Relay Attacks	All versions	N/A	Credential relay exploitation

SMB Exploitation Attack Chain

Common SMB attack progression:

SMB Attack Flow
1. Discover SMB Service ⇒ Port 139 or 445 open
2. Enumerate Shares and Users ⇒ Null session or null credentials
3. Check for Vulnerabilities ⇒ SMB version identification
4. Exploitation Method Selection ⇒ EternalBlue, relay, or brute force
5. Gain Access ⇒ RCE, file access, or credentials
6. Post-Exploitation ⇒ Lateral movement, persistence

7.3.2 SMB Enumeration Tools

Enum4Linux

Comprehensive SMB enumeration tool.

[EXAMPLE] Enum4Linux Usage

```
# Full enumeration
enum4linux 192.168.1.100

# Extract specific information
enum4linux -a 192.168.1.100  # All options
enum4linux -U 192.168.1.100  # User enumeration
enum4linux -S 192.168.1.100  # Share enumeration
enum4linux -N 192.168.1.100  # NetBIOS names
enum4linux -P 192.168.1.100  # Password policy
```

[OUTPUT] Terminal Output: Enum4Linux Output

```
root@kali:~# enum4linux 192.168.1.100

Starting enum4linux v0.8.9 on Mon Jan 12 14:00:00 2024

|    Target Information    |
==========================
Target ........... 192.168.1.100
RID Range ........ 500-550,1000-1050
Username ......... ''
Password ......... ''

===== Enumerating Workgroup/Domain on 192.168.1.100 =====
[+] Got domain/workgroup name: WORKGROUP

===== Enumerating Users on 192.168.1.100 =====
[+] Enumerating users
User:[admin] RID: [500]
User:[guest] RID: [501]
User:[john] RID: [1000]
User:[sarah] RID: [1001]

===== Enumerating Shares on 192.168.1.100 =====
Share:[IPC$] Type: STYPE_IPC_HIDDEN
Share:[Documents] Type: STYPE_DISKTREE
Share:[Backups] Type: STYPE_DISKTREE
Share:[Public] Type: STYPE_DISKTREE
```

SMBClient

Direct SMB client for Windows file shares.

[EXAMPLE] SMBClient Usage

```
# List shares
smbclient -L \\\\192.168.1.100 -N

# Connect to share
smbclient \\\\192.168.1.100\\Documents -N

# Download files
smb: \> get filename.txt

# Upload files
smb: \> put malware.exe

# Recursive download
smb: \> recurse ON
smb: \> mget *
```

SMBMap

Enumerate SMB shares and permissions.

[EXAMPLE] SMBMap Usage

```
# List shares
smbmap -H 192.168.1.100

# List shares with credentials
smbmap -H 192.168.1.100 -u admin -p password

# Recursive listing
smbmap -H 192.168.1.100 -R

# Download files
smbmap -H 192.168.1.100 -R --download "Backups/database.sql"
```

7.3.3 Exploiting SMB Vulnerabilities

Null Session Access

Some systems allow null sessions (no credentials) to enumerate information.

[EXAMPLE] Null Session Enumeration

```
# Connect with null session
smbclient \\\\192.168.1.100\\IPC$ -N

# Enumerate shares with null session
enum4linux -a 192.168.1.100

# Null session is extremely dangerous as it may reveal:
# - User accounts
# - Share names
# - Group information
# - Policy details
```

[WARNING] Null Session Mitigation

Disable null sessions by setting registry keys:

- Restrict Anonymous access to SAM database
- Configure "Restrict anonymous" policy

- Disable IPC*share*

SMB Relay Attacks

See detailed section below on SMB Relay.

7.3.4 NetBIOS Enumeration

NBTStat

Windows tool for NetBIOS name information.

[EXAMPLE] NBTStat on Linux (using nmblookup)

```
# Query NetBIOS names
nmblookup -A 192.168.1.100

# Broadcast lookup
nmblookup -B 192.168.1.255 \*

# Workgroup enumeration
nmap -sU -p 137 192.168.1.0/24
```

[OUTPUT] Terminal Output: NBT Scan Results

```
root@kali:~# nmblookup -A 192.168.1.100

Looking up status of 192.168.1.100
        DESKTOP-ABC1234         <00> -         B <ACTIVE>
        WORKGROUP               <00> - <GROUP> B <ACTIVE>
        DESKTOP-ABC1234         <20> -         B <ACTIVE>
        WORKGROUP               <1e> - <GROUP> B <ACTIVE>
        WORKGROUP               <1d> -         B <ACTIVE>
        ..__MSBROWSE__          <01> - <GROUP> B <ACTIVE>

MAC Address = AA:BB:CC:DD:EE:FF
```

7.4 SNMP Enumeration

SNMP (Simple Network Management Protocol) is used for network device management and monitoring. SNMP can leak valuable network information.

7.4.1 SNMP Fundamentals

SNMP Basics

- **Port:** 161/UDP (agent), 162/UDP (trap)
- **Versions:** SNMPv1, SNMPv2c, SNMPv3
- **Community String:** Authentication mechanism (often weak or default)
- **OID (Object Identifier):** Path to specific network information

SNMP Information Disclosure Risks

Information Type	OID Range	Security Impact
System Description	1.3.6.1.2.1.1.1	Identifies OS, services, versions
System Uptime	1.3.6.1.2.1.1.3	Helps predict patch status
Network Interfaces	1.3.6.1.2.1.2	Reveals network topology
Running Processes	1.3.6.1.2.1.25.4	Service discovery
Network Routes	1.3.6.1.2.1.4.21	Network segmentation details
TCP Connections	1.3.6.1.2.1.6.13	Active services and connections
User Accounts	1.3.6.1.4.1.77.1.2.25	Windows-specific user enumeration

SNMP Attack Progression

Typical SNMP exploitation workflow:

SNMP Information Gathering Chain
1. Identify SNMP Service (Port 161) ⇒ UDP port scan
2. Guess Community Strings ⇒ Try: public, private, community, internal
3. Enumerate System Information ⇒ OS type, version, uptime
4. Enumerate Network Details ⇒ Interfaces, routing, active connections
5. Extract Services and Processes ⇒ Identify vulnerable services
6. Correlate with Known CVEs ⇒ Develop exploitation strategy

Default SNMP Community Strings

Common default community strings (easily guessable):

- **public:** World-readable (default for SNMPv1/v2c)
- **private:** World-writable (dangerous)
- **community:** Alternative default
- **internal:** Corporate networks

7.4.2 SNMP Enumeration Tools

SNMPWalk

Recursively enumerate all SNMP objects.

[EXAMPLE] SNMPWalk Usage

```
# Enumerate with public community string
snmpwalk -c public -v 1 192.168.1.100

# Enumerate specific OID (system information)
snmpwalk -c public -v 1 192.168.1.100 1.3.6.1.2.1.1

# Enumerate sysDescr (system description)
snmpwalk -c public -v 1 192.168.1.100 1.3.6.1.2.1.1.1.0

# Enumerate sysObjectID (system type)
snmpwalk -c public -v 1 192.168.1.100 1.3.6.1.2.1.1.2.0

# Enumerate network interfaces
snmpwalk -c public -v 1 192.168.1.100 1.3.6.1.2.1.2
```

[OUTPUT] Terminal Output: SNMPWalk Output

```
root@kali:~# snmpwalk -c public -v 1 192.168.1.1

SNMPv2-MIB::sysDescr.0 = STRING: Cisco IOS Software, C2960X Software
SNMPv2-MIB::sysObjectID.0 = OID: SNMPv2-SMI::enterprises.9.9.46.1
SNMPv2-MIB::sysUpTime.0 = Timeticks: (5241900) 14 days, 14:36:40.00
SNMPv2-MIB::sysContact.0 = STRING: admin@company.com
SNMPv2-MIB::sysName.0 = STRING: switch.company.local
SNMPv2-MIB::sysLocation.0 = STRING: Building 5, Floor 3
SNMPv2-MIB::sysServices.0 = INTEGER: 78
```

SNMPGet

Query specific SNMP objects.

[EXAMPLE] SNMPGet Usage

```
# Get system name
snmpget -c public -v 1 192.168.1.1 1.3.6.1.2.1.1.5.0

# Get system uptime
snmpget -c public -v 1 192.168.1.1 1.3.6.1.2.1.1.3.0

# Get system description
snmpget -c public -v 1 192.168.1.1 1.3.6.1.2.1.1.1.0
```

7.4.3 Common SNMP OIDs

OID	Description	Use
1.3.6.1.2.1.1.1.0	sysDescr	System description
1.3.6.1.2.1.1.3.0	sysUptime	Uptime information
1.3.6.1.2.1.1.5.0	sysName	Hostname
1.3.6.1.2.1.1.6.0	sysLocation	Physical location
1.3.6.1.2.1.1.9	sysServices	Services running
1.3.6.1.2.1.25.1.1.0	hrSystemUptime	Host resources uptime
1.3.6.1.2.1.25.3.2	hrDeviceDescr	Device descriptions
1.3.6.1.2.1.25.4.2.1.2	hrSWRunName	Running processes

7.4.4 Extracting Information via SNMP

System Enumeration

[EXAMPLE] Comprehensive System Information

```
# Extract system name, OS, and uptime
snmpget -c public -v 1 192.168.1.100 \
  1.3.6.1.2.1.1.1.0 \
  1.3.6.1.2.1.1.5.0 \
  1.3.6.1.2.1.1.3.0

# Get all SNMP information at once
snmpwalk -c public -v 1 192.168.1.100 > snmp_dump.txt
```

Process Enumeration

[EXAMPLE] List Running Processes via SNMP

```
# Enumerate running processes (if available)
snmpwalk -c public -v 1 192.168.1.100 1.3.6.1.2.1.25.4.2.1.2

# Reveals applications and services running on the device
```

[WARNING] SNMP Security

SNMP community strings are often weak or default. This is a critical security issue:

- Change default community strings
- Use SNMPv3 with authentication and encryption
- Restrict SNMP access with ACLs
- Disable SNMP if not needed

7.5 SMB Relay Attack

SMB Relay is an active attack that intercepts and relays NTLM authentication requests to compromise systems without cracking passwords.

7.5.1 Attack Overview

How SMB Relay Works

1. **Interception:** Attacker intercepts NTLM authentication request
2. **Relay:** Authentication is forwarded to target system
3. **Impersonation:** Target system accepts forwarded credentials
4. **Access:** Attacker gains access with victim's privileges

Prerequisites for SMB Relay

- **Network Position:** Attacker on same network or can intercept traffic
- **SMB Authentication:** Target must use SMB for authentication
- **No SMB Signing:** Signatures prevent relay (must be disabled)
- **No EPA/Extended Protection:** Extended protection prevents relay
- **Different Services:** Authentication from Service A relayed to Service B

7.5.2 Reconnaissance for SMB Relay

Identifying Signing Status

[EXAMPLE] Check SMB Signing

```
# Using nmap
nmap --script smb-security-mode.nse -p 445 192.168.1.100

# Using Crackmapexec
crackmapexec smb 192.168.1.0/24

# Check signing with smbclient
smbclient -L \\\\192.168.1.100 -N
```

[OUTPUT] Terminal Output: Nmap SMB Signing Check

```
PORT     STATE SERVICE
445/tcp open  microsoft-ds

Host script results:
| smb-security-mode:
|   Account used for smb scripts: guest
|   User name: (null)
|   Password: (null)
|   Password required: false
|   Message signing disabled!
|   Message signing enabled but not required
|_  Only administrators are allowed to login with null session
```

7.5.3 Executing SMB Relay Attack

Using Responder and Ntlmrelayx

The classic SMB Relay attack chain:

[EXAMPLE] SMB Relay Attack Steps

```
# Step 1: Start Responder (capture NTLM auth attempts)
responder -I eth0 -rdw

# Step 2: In another terminal, start ntlmrelayx
ntlmrelayx.py -t 192.168.1.100 -c "whoami"

# Step 3: Trigger authentication (e.g., via phishing link)
# Victim browses to attacker's IP
# \\192.168.1.50\share (triggers NTLM authentication)

# Step 4: Responder captures auth, ntlmrelayx relays it
# Commands execute with victim's privileges
```

[OUTPUT] Terminal Output: Responder Capture

```
[*] [LLMNR]  192.168.1.100 has queried for nonexistent name
[*] [BROWSER] Sending BROWSER IsWorkGroupMaster response
[+] [LLMNR] Sending LLMNR response for nonexistent name
[+] [LLMNR] Sent LLMNR response to 192.168.1.100
[+] [BROWSER] Sent BROWSER response
[NTLMv2] Complete hash is: DOMAIN\USER::DOMAIN::xxx...

[-] Saving NTLM hashes to ntlmv2-ssl.txt
```

Interactive SMB Relay

[EXAMPLE] Ntlmrelayx Interactive Session

```
# Start for interactive shell
ntlmrelayx.py -t 192.168.1.100 -i

# This opens a prompt to interact with SMB
# ntlmrelayx>
# Can execute commands, upload files, etc.
```

7.5.4 Mitigating SMB Relay

Defense Mechanisms

[TIP] SMB Relay Prevention

Enable SMB Signing:

- **Windows Group Policy:**
  - Computer Configuration -> Policies -> Windows Settings -> Security Settings -> Local Policies -> Security Options
  - "Microsoft network server: Digitally sign communications (if client agrees)" = Enabled
  - "Microsoft network server: Digitally sign communications (always)" = Enabled
- **Samba:** Set "server signing = mandatory" in smb.conf

Additional Protections:

- Enable Extended Protection for Authentication (EPA)
- Implement multi-factor authentication
- Network segmentation (isolate critical systems)
- Monitor for NTLM authentication attempts

7.5.5 Detection

Detecting SMB Relay Attacks

[EXAMPLE] Log Analysis

```
# Look for NTLM authentication in Windows Event Viewer
# Event ID 4625 (Failed login)
# Event ID 4624 (Successful login)

# Check for suspicious lateral movement
# Logins from unusual IPs or at unusual times
```

[WARNING] SMB Relay Impact

SMB Relay can be devastating:

- No password cracking needed
- Leverages legitimate NTLM authentication
- Can escalate privileges
- Difficult to detect without proper logging
- Can compromise entire domains

Always enable SMB signing on production systems!

8 Summary

This section covered critical network attack techniques:

- **Wireshark:** Capturing and analyzing network traffic
- **Network Enumeration:** Discovering hosts and services
- **SMB/NetBIOS:** Enumerating file shares and user accounts
- **SNMP:** Extracting network device information
- **SMB Relay:** Relaying NTLM authentication for unauthorized access

These techniques form the foundation of network-level attacks and are essential knowledge for the eJPTv2 certification exam.

[WARNING] Ethical Considerations

All enumeration and attack techniques described here must only be used in authorized environments:

- Only test systems you have explicit written permission to attack
- Respect privacy and legal boundaries
- Report vulnerabilities responsibly
- Use practice environments (HackTheBox, TryHackMe)

Unauthorized network attacks are illegal and unethical.

9 Practical Labs & Practice Environment

To master network attacks, hands-on practice in controlled environments is essential.

9.1 Practical Lab Exercises

9.1.1 Hands-On Practice Scenarios

The following scenarios help reinforce the network attack techniques covered in this section. Practice these attacks in controlled environments with proper authorization.

Exercise 1: Wireshark Traffic Analysis

[SCENARIO] Practice: Capture and Analyze Network Traffic

Learning Objectives:

- Understand packet structure and network layers
- Apply display and capture filters effectively
- Extract sensitive data from unencrypted traffic
- Recognize plaintext protocol risks

Key Skills:

1. Capture network traffic on specific interfaces
2. Filter for HTTP, FTP, DNS, and SMB traffic
3. Follow TCP streams to reconstruct conversations
4. Analyze packet headers and payload contents
5. Extract credentials from captured packets

Expected Outcomes:

- Identify credential transmission in plaintext protocols
- Understand protocol vulnerabilities through packet inspection
- Develop systematic network analysis methodologies

Exercise 2: Network Enumeration Techniques

[SCENARIO] Practice: Discover Services and Vulnerabilities

Learning Objectives:

- Perform systematic network reconnaissance
- Identify active hosts and open services
- Discover service versions and configurations
- Map network topology and relationships

Key Skills:

1. Use Nmap for comprehensive port scanning
2. Perform host discovery with ARP and ICMP
3. Enumerate DNS and service information
4. Analyze service banners and version information
5. Document findings methodically

Expected Outcomes:

- Create detailed network maps
- Identify exploitable services
- Document reconnaissance findings

Exercise 3: SMB and File Sharing Analysis

[SCENARIO] Practice: Discover Shares and Extract Information

Learning Objectives:

- Understand SMB protocol vulnerabilities
- Enumerate users, groups, and shares
- Identify misconfigured permissions
- Extract sensitive data from file shares

Key Skills:

1. Run enum4linux for comprehensive SMB enumeration
2. Use smbclient to interact with shares
3. Analyze SMB session security and signing
4. Identify relay attack opportunities
5. Understand SMB version implications

Expected Outcomes:

- Successfully list accessible shares
- Extract configuration files and credentials
- Identify SMB vulnerabilities and misconfigurations

Exercise 4: SNMP Information Gathering

[SCENARIO] Practice: Extract SNMP Information Disclosure

Learning Objectives:

- Understand SNMP protocol weaknesses
- Enumerate system and network information via SNMP
- Identify default community strings
- Extract sensitive configuration details

Key Skills:

1. Query SNMP services with snmpwalk
2. Extract system information from MIB
3. Enumerate network interfaces
4. Identify running processes and services
5. Discover SNMP community strings

Expected Outcomes:

- Successfully enumerate system details via SNMP
- Map network topology using SNMP data
- Understand SNMP vulnerability implications

Exercise 5: DNS Enumeration and Zone Transfers

[SCENARIO] Practice: Perform DNS Reconnaissance

Learning Objectives:

- Understand DNS enumeration techniques
- Attempt zone transfers for subdomain discovery
- Enumerate DNS records and resolve names
- Map network infrastructure via DNS

Key Skills:

1. Query nameservers using dig and nslookup
2. Attempt zone transfers (AXFR)
3. Enumerate DNS records (A, MX, TXT, NS)
4. Perform DNS brute-forcing
5. Analyze DNS response patterns

Expected Outcomes:

- Discover subdomains and virtual hosts
- Extract DNS infrastructure information
- Identify misconfigured DNS servers

9.2 Recommended Online Lab Platforms

Instead of local Docker setups, practice these techniques using professional-grade lab environments:

9.2.1 HackTheBox & TryHackMe Machines

Platform	Machine Name	Focus Area
HackTheBox	Included (Easy)	Wireshark, packet analysis, plaintext protocols
HackTheBox	Lame (Easy)	SMB enumeration, Samba service exploitation
HackTheBox	Legacy (Easy)	EternalBlue, SMB-based network attacks
TryHackMe	Network Services	Comprehensive SMB, FTP, SSH enumeration
TryHackMe	Wireshark 101	Traffic analysis fundamentals and packet inspection
TryHackMe	SNMP Security	SNMP enumeration and information disclosure

Why Use Online Platforms?

- **Designed for Learning:** Machines are specifically configured to teach vulnerable services
- **Official Support:** Walkthroughs and hints available for stuck students
- **Community:** Active forums and discussion with other learners
- **Realistic Environment:** More authentic than simplified Docker scenarios
- **No Local Setup:** No Docker installation or maintenance required
- **Progression Path:** Curated difficulty progression from beginner to advanced
- **Certificates:** Completion badges and certifications to demonstrate skills

9.3 Best Practices for Lab Practice

Systematic Approach

[EXAMPLE] Lab Methodology

For each practice machine, follow this structured approach:

1. **Reconnaissance:** Gather information about the target
   - Use Nmap for port and service discovery
   - Identify running services and versions
   - Document all findings
2. **Enumeration:** Deep dive into discovered services
   - Enumerate users and groups
   - Discover file shares and permissions
   - Extract system information
   - Identify configuration details
3. **Analysis:** Assess discovered vulnerabilities
   - Research service vulnerabilities
   - Identify misconfigurations
   - Determine exploitation paths
   - Assess business risk
4. **Documentation:** Create detailed reports
   - Screenshot key findings
   - Annotate important details
   - Document commands and outputs
   - Include remediation recommendations

Documenting Your Findings

[EXAMPLE] Lab Report Template

For each attack scenario, document:

1. **Target Information:** IP address, hostname, scope
2. **Reconnaissance Results:** Services, versions, OS
3. **Enumeration Findings:** Users, shares, configurations
4. **Vulnerabilities Discovered:** List all identified weaknesses
5. **Exploitation Proof:** Screenshots and evidence of access
6. **Impact Assessment:** Business impact of vulnerabilities
7. **Remediation Steps:** How to fix the issues
8. **Key Learnings:** What this exercise taught you

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